Exploring the Role of Generative AI and Deep Learning in Personalized Education Content: a Systematic Literature Review

Abstract

The rise of generative artificial intelligence (AI) and deep learning has introduced new possibilities for personalized education. However, existing research lacks a comprehensive synthesis of how these technologies contribute to the design and delivery of personalized learning content.This study presents a systematic literature review (SLR) examining the role of generative AI and deep learning in personalized education. Following PRISMA 2020 guidelines, 39 peer-reviewed articles published between 2022 and 2025 were selected from databases including Scopus, Web of Science, IEEE Xplore, ERIC, and ScienceDirect. Thematic analysis was conducted using NVivo, and the quality of included studies was assessed using the CASP checklist. Three major themes emerged: (1) generative AI enables adaptive content creation and real-time instructional support; (2) deep learning enhances learner profiling and predictive feedback through multimodal data analysis; and (3) ethical, cultural, and equity-related challenges persist, including concerns about algorithmic bias, data privacy, and teacher displacement. While these technologies show promise in scaling personalization and improving engagement, issues related to trust, transparency, and contextual relevance remain significant. Generative AI and deep learning offer transformative potential for personalized learning, yet their successful implementation depends on human oversight, ethical safeguards, and alignment

with pedagogical goals. Future research should focus on long-term learning outcomes, cross-cultural validation, and the development of explainable and inclusive AI systems.